

Jump Modeling with Dirichlet Process Mixture

Yuru Sun¹ Worapree (Ole) Maneesoonthorn¹ Yong Song² Wei Wei¹

¹Monash University ²University of Melbourne

M

Motivation

Return jumps - defined as discontinuous, sudden and extreme changes, adds to the **heavy-tail** feature of return distribution and helps to explain the empirical feature of the **option implied volatility** smile. Traditionally, a parametric jump model is used. We propose a more flexible method for modeling jumps using Dirichlet Process Mixture (DPM).

Parametric Model:

- Assume jump size follows a specific distribution: "Normal", "Exponential", "Double-exponential",
- Not flexible enough if considering both positive and negative jumps or prediction
- but it's easy to estimate and bring convenience for pricing options

DPM Jump Model

DPM Jump Model:

- Assume jump size follows a **flexible mixture** of distributions,
- Reveal **data-driven** jump size distribution,
- Capability of better capturing both positive and negative jumps.

Let r_t be the daily log return. Let h_t be the latent log variance, ΔN_t be the jump occurrence and Z_t be the jump size. Our proposed model follows:

Measurement Equations:

$$r_t = \exp(h_t/2)\epsilon_t^r + \Delta N_t Z_t, \quad (1)$$

State Equations:

$$h_t = \phi_0 + \phi_1 h_{t-1} + \sigma_h \epsilon_t^h, \quad \Delta N_t \sim \text{Bernoulli}(\delta) \quad (2)$$

$$Z_t \sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2), \quad (3)$$

where

$$w_k = \nu_k \prod_{l=1}^{k-1} (1 - \nu_l), \quad \nu_k \sim \text{Beta}(1, \alpha). \quad (4)$$

We assume $(\epsilon_t^r, \epsilon_t^{BV}, \epsilon_t^h)$ follow $N(0, 1)$.

Sampling Mechanics

A. How to deal with the infinite sum - Slice Sampler:

Introducing an auxiliary variable u_t who is uniformly distributed conditional on $\mathbf{w}$, then we have

$$p(Z_t, u_t | \cdot) \propto \sum_{k=1}^{\infty} \frac{1}{w_k} I(u_t < w_k) w_k N(Z_t | \mu_k, \sigma_k^2), \quad (5)$$

$$\propto \sum_{k \in A_t} N(Z_t | \mu_k, \sigma_k^2),$$

where $A_t = \{k : u_t < w_k\}$. Note here A_t is a **finite** set where the condition is satisfied.

B. How we deal with the non-linear state-space model - "7 component mixture":

$$r_t^* = r_t - \Delta N_t Z_t, \quad \log(r_t^{*2} + 0.001) = h_t + \sum_{i=1}^7 q_i N(m_i - 1.2704, \nu_i^2), \quad (6)$$

We approximate the distribution of $\log(\epsilon_t^{r*2})$ as a mixture of seven component with pre-selected mean m_i and variance ν_i^2 in the literature.

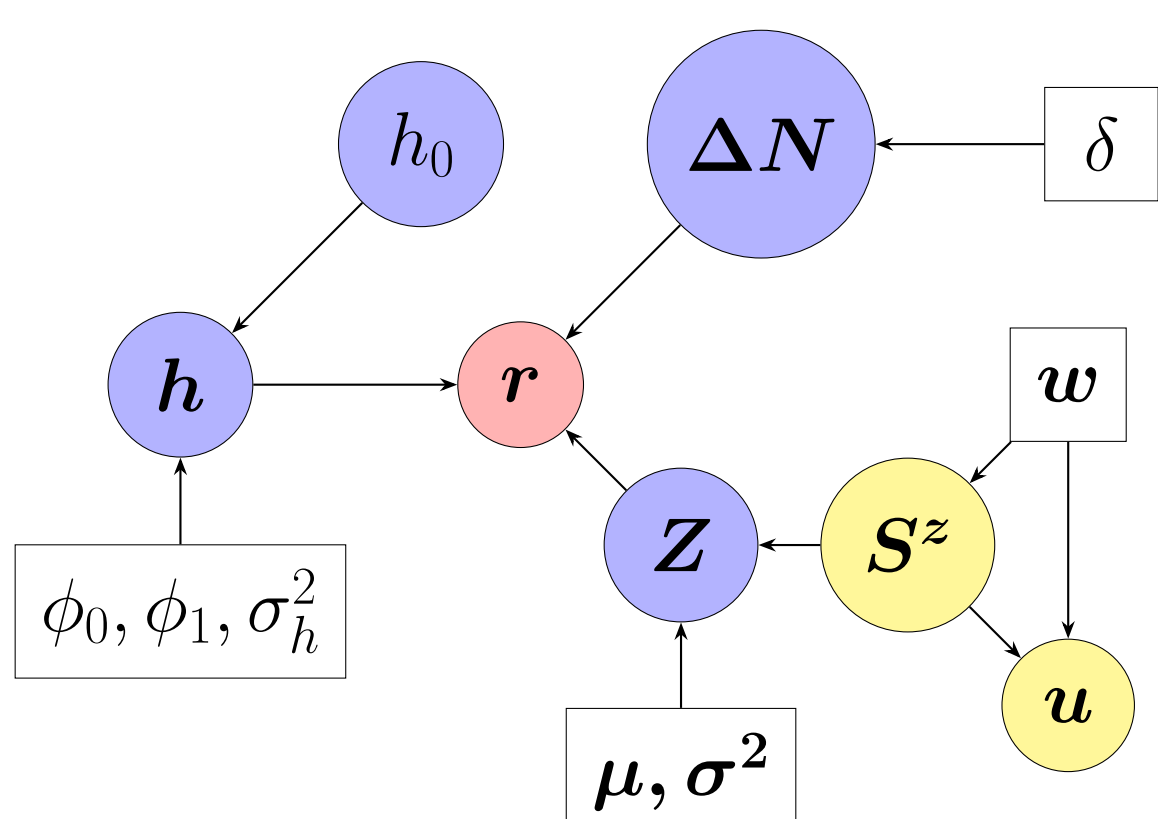


Figure 1. DAGs of the DPM jump model structure. Red circles represents observed data, purple circles are latent variables. Parameters are represented as rectangular. Yellow circles represents the auxiliary variables where $\mathbf{u}$ is introduced for slice sampler and $\mathbf{S}^z$ is an indicator functions for components.

Results Discussion

For **simulation**, under different data generating process (DGP), we investigate the model performance on the:

- Proportion of correctly identify jumps (Sensitivity) and correctly identified non-jumps (Specificity);
- Estimated posterior jump kernels $\tilde{\mathbf{J}}$.

For **empirical studies**, with different financial indices, we investigate:

- Estimated jump intensity;
- Comparison with Barndorff-Nielsen and Shephard (BNS) jump hypothesis test at 1% significance level;
- Estimated posterior jump kernels $\tilde{\mathbf{J}}$.

BNS jump test:

$$BNS_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{(\pi^2/4 + \pi - 5)N^{-1}\max(1, TP_t/BV_t^2)}}, \quad (7)$$

where

$$TP_t = [2^{2/3}\Gamma(7/6)\Gamma(1/2)^{-1}]^{-3} \left(\frac{N^2}{N-2} \right) \sum_{i=3}^N |r_{t-i-2}|^{4/3} |r_{t-i-1}|^{4/3} |r_{t-i}|^{4/3}, \quad (8)$$

$$RV_t = \sum_{i=1}^N |r_{t-i}|^2, \quad BV_t = \frac{\pi}{2} \left(\frac{N}{N-1} \right) \sum_{i=2}^N |r_{t-i}| |r_{t-i+1}|.$$

r_{t_i} is the i th of N observed returns between trading day t and $t+1$.

Estimated posterior jump kernel $\tilde{\mathbf{J}}$:

For total M MCMC iterations,

$$p(\Delta N_t = 1) = \frac{1}{M} \sum_{i=1}^M \Delta N_t^{(i)}, \quad \tilde{Z}_t = \frac{1}{M} \sum_{i=1}^M Z_t^{(i)}, \quad \tilde{J}_t = \tilde{Z}_t | p(\Delta N_t = 1) > 0.5. \quad (9)$$

Simulation Study

We simulate data assuming the true DGP follows Equations (1) and (2), and the true jump size follows a mixture of two truncated normals to mimic the extreme values of jumps in practice. $Z_t \sim 0.6TN(-\mu_z, 0.8, -\infty, -\mu_z) + 0.4TN(\mu_z, 0.6, \mu_z, +\infty)$. Our simulation results show the model performance under different true DGPs.

Jump Size Sensitivity Analysis			
	2- σ	2.5- σ	3- σ
Sensitivity			
Parametric	0.6375 (0.5882, 0.6845)	0.7838 (0.7576, 0.8131)	0.8296 (0.8021, 0.8610)
DPM	0.6114 (0.5722, 0.6524)	0.7587 (0.7323, 0.7879)	0.8202 (0.7941, 0.8503)
Specificity			
Parametric	0.8855 (0.8635, 0.9060)	0.8845 (0.8663, 0.9012)	0.8971 (0.8811, 0.9117)
DPM	0.9469 (0.9313, 0.9619)	0.9634 (0.9553, 0.9711)	0.9690 (0.9614, 0.9752)

Table 1. Performance of parametric and DPM models as the **true jump size changes**. The true jump intensity $\delta = 10\%$ and μ_z changes from 2- σ to 3- σ , away from the conditional mean of return. The post-convergence average and 90% posterior interval of correctly identified jump proportion (sensitivity) and the correctly identified no-jump proportion (specificity) are provided in the table.

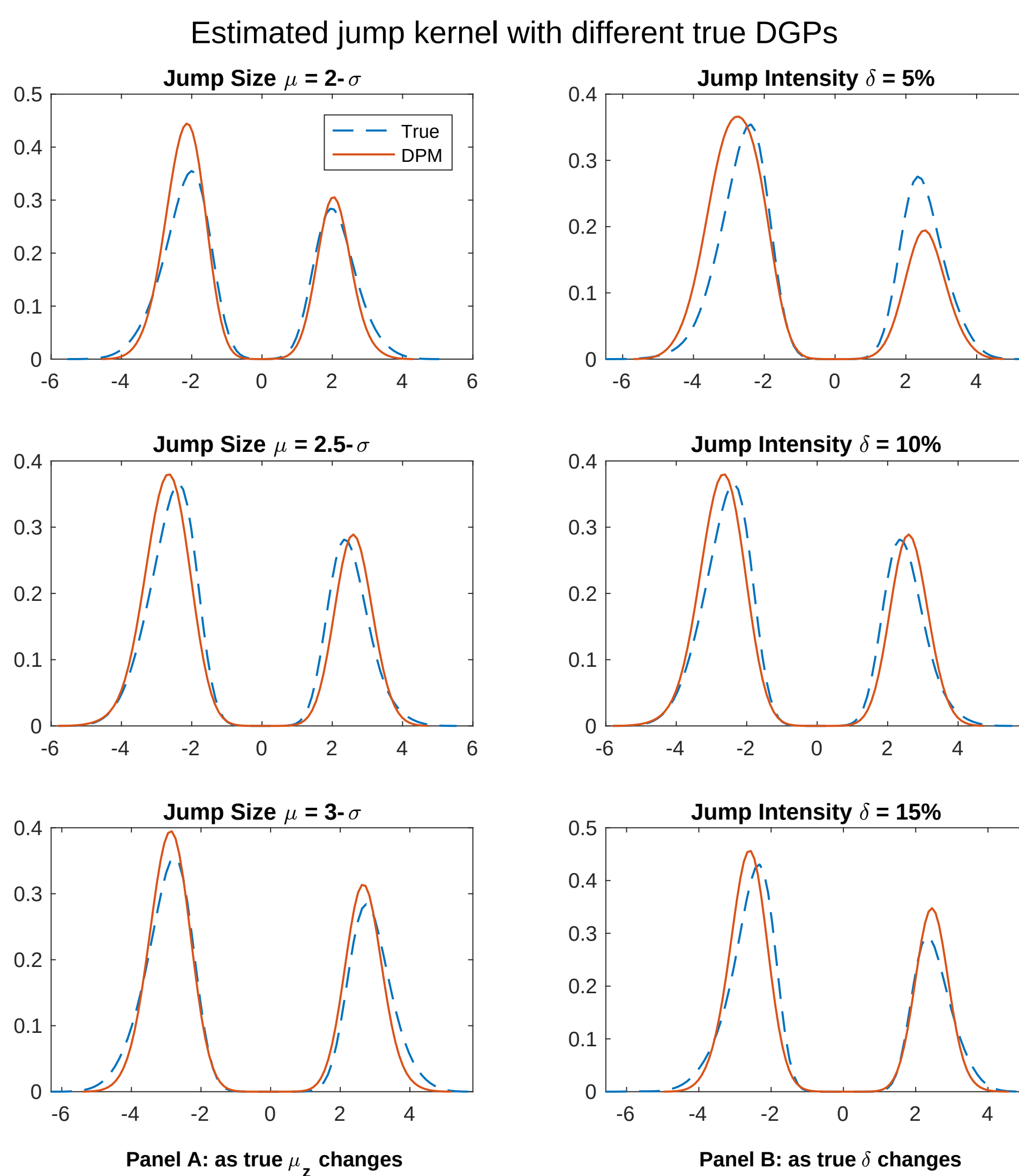


Figure 2. The **estimated jump kernel** for both parametric and DPM jump models under **different true DGPs**, as stated in the "Results Discussion" section. The blue dashed line represents the true jump density. **Panel A (left column): true jump size changes** from 2- σ to 3- σ away from return's conditional mean, while the jump intensity fixed at 10%. Panel A corresponds to Table 1. **Panel B (right column): the true jump intensity δ changes** from 5% to 15% while the true jump size fixed at 2.5- σ away from the conditional mean. Panel B corresponds to Table 1.

Jump Intensity Sensitivity Analysis

	5%	10%	15%
Sensitivity			
Parametric	0.8316 (0.7963, 0.8704)	0.7838 (0.7576, 0.8131)	0.8374 (0.8142, 0.8594)
DPM	0.8230 (0.7870, 0.8565)	0.7587 (0.7323, 0.7879)	0.8131 (0.7917, 0.8351)
Specificity			
Parametric	0.9344 (0.9218, 0.9456)	0.8845 (0.8663, 0.9012)	0.8471 (0.8300, 0.8642)
DPM	0.9725 (0.9643, 0.9802)	0.9634 (0.9553, 0.9711)	0.9578 (0.9504, 0.9658)

Table 2. Performance of parametric and DPM models as the **true jump intensity changes**. The true jump size is fixed at 2.5- σ away from the conditional mean of return. The true jump intensity changes from 5% to 15%. The post-convergence average and 90% posterior interval of correctly identified jump proportion (sensitivity) and the correctly identified no-jump proportion (specificity) are provided in the table.

Empirical Study

Estimated Jump Intensity for Financial Indices				
	SPX	XOI	TYX	XAU
Parametric	0.0918 (0.0673, 0.1225)	0.1313 (0.0969, 0.1710)	0.1398 (0.1040, 0.1799)	0.0529 (0.0392, 0.0680)
DPM	0.0894 (0.0622, 0.1182)	0.0961 (0.0666, 0.1291)	0.1334 (0.0946, 0.1739)	0.1191 (0.0933, 0.1484)
BNS test (1%)	0.0741	0.0676	0.1427	0.0438

Table 3. Post-convergence **posterior mean of jump intensity δ** and their 90% posterior interval. The bottom row shows the Barndorff-Nielsen and Shephard (BNS) jump hypothesis test at 1% significance level. The bolded numbers are the estimated jump intensity that is closer to the BNS jump test results.

Estimated Jump Kernel

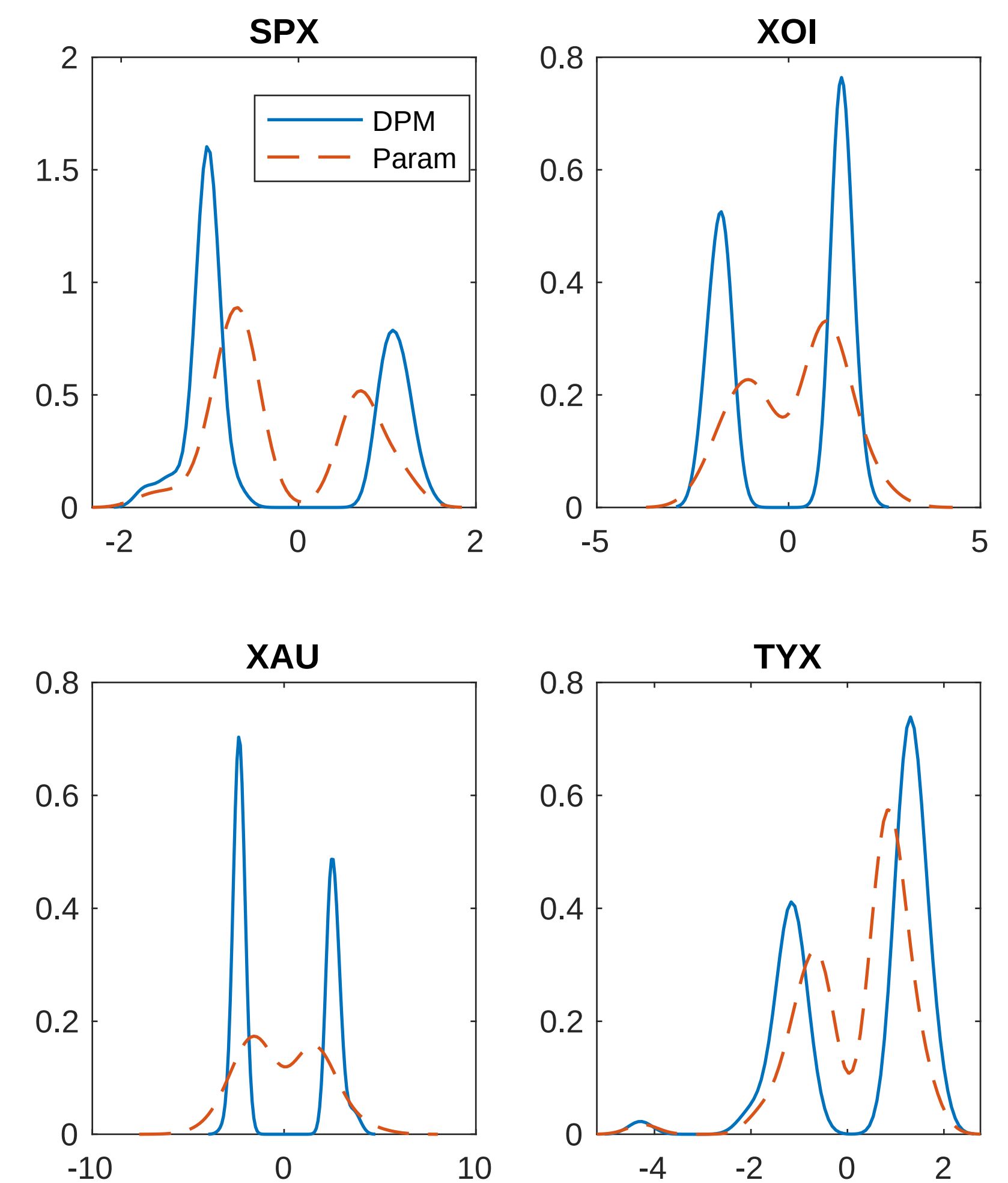


Figure 3. The **estimated jump kernel** of both parametric and DPM jump models for four financial indices, which is stated in "Results Discussion" Section. The blue solid line represents the jump kernel estimated from DPM jump model. The red dashed line represents the estimated jump kernel from parametric jump model.

Highlights of Contributions

- Higher specificity when jump size changes:** from our simulation results, our DPM jump model has a lower chance of identifying a false jump than parametric model, with a slight trade-off on sensitivity. As the true jump size becomes bigger, both parametric and DPM jump model better identify jumps.
- More distinguished modes of estimated jump kernel:** from both simulation results and empirical results, compared with the parametric model, the estimated jump kernel from DPM jump model has lower mass around zero, which would usually be classified as no-jumps, which indicates that the DPM jump model is more capable of distinguishing jumps from the latent log volatility.
- Estimated jump intensity might depends on the series we use:** from the empirical analysis, for SPX and XOI, the DPM model estimated jump intensity is closer to the BNS test, and for XAU and TYX, the parametric model seems obtain a results that is closer to BNS test.